

Andrew Pols

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Education

University of Toronto

Sep. 2024 - Present

- HBSc., **Computer Science** | **GPA: 4.00 / 4.00**
- Relevant coursework: Software Design; Computer Organization; Software Tools and Systems Programming

Experience

Software Developer Intern

May 2026 - May 2027

IBM Canada Ltd.

- Developed a **dependency graph visualization** for **IBM Planning Analytics Workspace**, enabling users to explore relationships between Planning Analytics objects through real-time graph construction and rendering.
- Engineered a dependency analysis pipeline using **TM1 REST APIs**, recursively traversing object relationships and parsing scripts to construct dependency graphs while accounting for optimization by lazy-loading data retrieval.
- Built production-ready front-end functionality with **React, TypeScript, and React Flow**, implementing graph layouts, object-specific interactions, filtering, and scalable visualization of complex dependency structures.
- Collaborated with **design, product management, QA, and engineering** to iterate on requirements, resolve technical constraints, and deliver client-facing functionality through an Agile development process.

Projects

ScholarSearch | [Source Code](#)

ReactJS | Python | PostgreSQL | FastAPI

- Collaborated in a 3-person team to build a research paper search engine that accelerates the research process by providing efficient searches and citation graphs with D3.js to view the impact a paper has.
- Stored papers in a PostgreSQL database, along with vector embeddings in ChromaDB for scalable semantic search.
- Implemented Redis caching on a database of over **1,000,000** papers to achieve a **sub-200ms** search response.
- Containerized Postgres, ChromaDB, and Redis systems with Docker Compose for a streamlined development cycle.

FindMySound | [Source Code](#)

ReactJS | Python | Django | Scikit-Learn

- Developed a web app that simplifies the music recommendation process using Spotify's Web API with **OAuth 2.0**.
- Recommended music with **95% satisfaction** through the K-Nearest-Neighbours ML algorithm with Scikit-Learn.
- Maintained a user base through a custom user model, with authentication through Django's Simple JWT.
- Created RESTful API endpoints to handle authentication, Spotify account syncing, and recommendations.

GroupFlow | [Source Code](#)

Java | MongoDB | Swing

- Programmed a Java project management desktop app that enhanced productivity of university students by providing a centralized platform for scheduling meetings and assigning tasks.
- Utilized MongoDB for persisting data between users in a group, and Java Swing for the user interface.
- Practiced industry-standard coding practices by adhering to **Clean Architecture** and SOLID design principles, separating each use case into its own package, and following the dependency layering principle.
- Effectively collaborated in a team of 6 by enforcing GitHub workflows like CheckStyle for code quality, and required code reviews for pull requests, ensuring only production-standard code is pushed to main.

Technical Skills

- **Programming Languages:** Java, Python, C, JavaScript, TypeScript, SQL, HTML5, CSS
- **Frameworks & Libraries:** React, Node.js, Next.js, Django REST, Flask, React Flow, Scikit-learn, OpenCV
- **Developer Tools:** Git, Jira, Docker, AWS, Azure, Redis, PostgreSQL, MongoDB, Nginx